

Wildlife and Species Richness of Canada **Student Activity Sheets**

Name: _____

**Most of the answers to these questions can be found in the Wildlife and Species Richness of Canada background section.*

- 1) Now that we have determined how a species is put on a COSEWIC list and the seven levels of risk, let's look at that list and learn more about those species. Access the Canadian Wildlife Service Web site www.speciesatrisk.gc.ca and fill in the table below with at least one species that falls within each of the seven levels of risk.

Table 1: Species at Risk

Level of risk	species name	category of species
extinct		
extirpated		
endangered		
threatened		
special concern		
not at risk		
data deficient		

- 2) Using the following table, make a classified map of Species-at-Risk for Canada by giving each ecozone a classification number (Classification 1) using equal intervals for the number of species at risk within that ecozone.

The method to do this is as follows: if you decide to use 5 classes, divide the range (lowest to highest values) into 5 equal intervals.

For example, if the range was from 1 to 100, the highest value (100) divided by 5 is the size of each interval. The example intervals would therefore be: 1. less than 20, 2. less than 40, 3. less than 60, 4. less than 80, 5. less than or equal to 100.

What problems do you see with this example in classifying your map?

Table 2: Species at Risk by Ecozone Classification Number

Ecozone names	COSEWIC99	Classification 1	Classification 2
	# Species at Risk		
Arctic Cordillera	9		
Northern Arctic	12		
Southern Arctic	14		
Taiga Plains	12		
Taiga Shield	14		
Boreal Shield	63		
Atlantic Maritime	42		
Mixedwood Plains	128		
Boreal Plains	25		
Prairies	46		
Taiga Cordillera	6		
Boreal Cordillera	10		
Pacific Maritime	55		
Montane Cordillera	42		
Hudson Plains	8		

- 3) Now, reclassify the ecozones using different classification intervals (Classification 2). Try using a different range (e.g. 6 to 128) or change your intervals to better differentiate the ecozone classifications by using **uneven intervals**. This can be done by limiting your range. An example would be to use intervals of <10, <20, <40, <60, >=60. How does this change the appearance of your map?

- 4) From Table 3 make a line graph of each bird, mammal, amphibian/reptile and tree category across the ecozones. Determine which ecozone has the largest number of species for birds, mammals, reptile/amphibians, and trees. (Ask your teacher to provide you with graph paper). Use four colours of lines (one for each class) on the same graph.

Birds	Colour: _____	Highest count in: Ecozone _____
Mammals	Colour: _____	Highest count in: Ecozone _____
Amphibian/reptiles	Colour: _____	Highest count in: Ecozone _____
Trees	Colour: _____	Highest count in: Ecozone _____

- 5) Can you think of a reason why that particular ecozone is highest for a category?

- 6) Can you give a reason why one ecozone is not the highest for all categories?

Table 3: Terrestrial Species within each of the Ecozones

Ecozone names	# of Bird Species	# of Mammal Species	# of Reptile/ Amphibian Species	# of Tree Species	Total # of Wildlife Species	Total minus COSEWIC Species
Arctic Cordillera	69	29	1	6		96
Northern Arctic	72	23	0	0		83
Southern Arctic	141	44	1	12		184
Taiga Plains	212	57	7	20		284
Taiga Shield	193	53	9	23		264
Boreal Shield	267	71	43	80		398
Atlantic Maritime	195	54	38	64		309
Mixedwood Plains	238	58	51	106		325
Boreal Plains	278	81	21	49		404
Prairies	280	84	26	44		388
Taiga Cordillera	174	47	1	13		229
Boreal Cordillera	194	52	6	28		270
Pacific Maritime	250	85	31	47		358
Montane Cordillera	265	92	33	52		400
Hudson Plains	185	50	9	26		262

7) Do you see a relationship between climate and number of species? (Explain)

8) Which category does climate affect the most?

9) Using the data from Table 3, complete the table by supplying the **total number of species** to determine the **Wildlife Species Richness**. (Calculate this by adding the bird, mammal, amphibian/reptile number of species. Leave out the tree species for now). Enter your values for each ecozone on a graph. Which ecozone has the highest?

10) Which has the lowest?

11) How does that profile compare with the number of trees in each ecozone?

12) Now map the total number of species (available from Table 3). Remember to add in the tree species. Note that this chart does not include all species in Canada, i.e. we have only used four categories and do not include fish, plants, etc. This chart is therefore only to be used as an approximation for example purposes.

- 13) Using the attribute table (Table 3) for the maps, create a new layer by using the number of species at risk subtracted from the total number of species. How do these two maps compare?

- 14) Locate the species richness map on the GeoGratis Web page by exploring the site at: <http://geogratis.cgdi.gc.ca/Ecosystem/ecosystem.html>. Compare your map to it.

- 15) Let's try some modeling to predict the effect of various factors on an ecozone's risk to species biodiversity. To simplify the calculations, we have classified our table in the same way you did above for Species at Risk.

Now, consider the relationship between population densities, road densities and the number of total species and COSEWIC species on biodiversity, that is, which factor would have the greatest impact on the risk to biodiversity if it were to change. Rank each of these factors from 1 (least) to 4 (most) as you think they affect the risk to biodiversity. Use this number to calculate the value in the appropriate columns below. The same factor is applied to each ecozone for a given category, e.g. if **species factor** is 3, then each ecozone value for **total species** is multiplied by 3.

Now, complete the mathematical equation for *weighted factor* below for each of the ecozones.

total species X species factor = species value

cosewic species X cosewic factor = cosewic value

pop.density X pop.factor = pop. value

road density X road factor = road value.

species value + cosewic value + pop. value + road value = **weighted value**

Table 4: Classified values for weighting factor calculation

Ecozone names	Total Species	Species Value	COSEWIC Species	COSEWIC Value	Population Density	Pop. Value	Road Density	Road Value	Weighted Value
Arctic Cordillera	2		1		1		1		
Northern Arctic	1		2		2		1		
Southern Arctic	2		2		2		1		
Taiga Plains	3		2		2		2		
Taiga Shield	3		2		2		1		
Boreal Shield	5		4		3		2		
Atlantic Maritime	4		3		4		3		
Mixedwood Plains	5		5		5		4		
Boreal Plains	4		3		3		3		
Prairies	4		3		4		3		
Taiga Cordillera	3		1		1		1		
Boreal Cordillera	3		1		2		2		
Pacific Maritime	4		4		4		2		
Montane Cordillera	4		3		3		2		
Hudson Plains	3		1		2		1		

- 16) Which ecozone is the greatest at risk by your calculation? _____

- 17) Is it what you would expect, and why?

18) Re-assess your factors and try again. How has this changed the risk to biodiversity of the ecozones?

19) What factor has the greatest effect on your model?
